Product Requirements & Specification Document

Project Name

LoanForest - Loan Default Predictor

Description

LoanForest is a Streamlit web application that predicts the risk of loan default using a Random Forest machine learning model. Users interact with the app via feature sliders to input loan and applicant details, receiving an instant prediction of default risk.

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1. Objectives

- Provide an intuitive interface for users to assess loan default risk.
- Leverage a trained Random Forest model for accurate predictions.
- Enable real-time, interactive user input and feedback.

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2. Stakeholders

Role	Responsibility
Product Owner	Feature definition, approval
Developers	Implementation, testing
Data Scientist	Model training, validation
End Users	Use app for risk assessment

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3. Functional Requirements

ID	Requirement
FR1	Users can input loan/applicant features via interactive sliders.
FR2	The app displays a predicted probability of loan default.
FR3	The app provides a clear, user-friendly interface.
FR4	The Random Forest model is pre-trained and loaded at app startup.
FR5	Input validation ensures all features are within valid ranges.
FR6	The app displays model confidence and a simple risk interpretation.
FR7	The app runs locally and is deployable to Streamlit Cloud.

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4. Non-Functional Requirements

ID	Requirement
NFR1	Response time < 2 seconds per prediction
NFR2	Codebase is modular and well-documented
NFR3	UI is responsive and accessible
NFR4	App dependencies are managed via requirements.txt

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5. User Interface Specifications

- **Main Page**
  - Title: “LoanForest - Loan Default Predictor”
  - Feature sliders for all model input features (e.g., income, loan amount, term, credit score)
  - “Predict” button
  - Output: Default risk probability, risk category (e.g., Low/Medium/High), and confidence
  - Minimalist, startup-themed design

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6. Model & Data Specifications

Aspect	Specification
Model Type	Random Forest Classifier (scikit-learn)
Input Features	Numeric/categorical features relevant to loans
Output	Probability of default (0-1), risk category
Model Storage	Serialized with `joblib` or `pickle`
Data Source	Public loan datasets (e.g., LendingClub)

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7. Technical Stack

Component	Technology
Frontend	Streamlit
Backend/ML	Python, scikit-learn
Deployment	Streamlit Cloud
Packaging	requirements.txt

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8. Implementation Outline

```python
# Pseudocode
import streamlit as st
import joblib

model = joblib.load('rf_model.pkl')
user_inputs = {...}  # Gathered from sliders
prediction = model.predict_proba([user_inputs])[0][1]
st.write(f"Default Risk: {prediction:.2%}")
```

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9. Acceptance Criteria

ID	Criteria
AC1	App loads and displays all input sliders
AC2	Prediction is generated within 2 seconds
AC3	Output includes probability and risk category
AC4	App runs locally and on Streamlit Cloud without errors
AC5	Code is documented and maintainable

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10. Out of Scope

- User authentication
- Model retraining via UI
- Database integration
- Advanced analytics or explainability features

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11. Milestones

Milestone	Target Date
UI Prototype	Week 1
Model Integration	Week 2
Testing & QA	Week 3
Deployment & Review	Week 4

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12. Dependencies

- Python 3.8+
- streamlit
- scikit-learn
- joblib

- pandas, numpy

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13. Risks & Mitigations

Risk	Mitigation
Poor model accuracy	Use robust public datasets, validate model
UI usability issues	User testing, iterative feedback
Dependency conflicts	Pin versions in requirements.txt

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14. Appendix

- Example datasets: LendingClub
- Streamlit docs: https://docs.streamlit.io/
- scikit-learn docs: https://scikit-learn.org/